Assignment No 3

Aim: Apply dimensionality reduction using PCA

Theory:

Dimensionality Reduction is a machine learning technique used to reduce the number of features or variables in a dataset while preserving the most important information present in the original data. A dataset with a large number of features is called **high-dimensional data**. Working with too many features can increase computational complexity, make visualization difficult, and sometimes reduce the performance of machine learning models.

Principal Component Analysis (PCA) is one of the most commonly used dimensionality reduction techniques. PCA is an **unsupervised learning technique** that transforms a large set of correlated variables into a smaller set of new variables called **principal components**.

The principal components are linear combinations of the original features. The first principal component captures the **maximum variance** in the data. The second principal component captures the maximum remaining variance while being independent (orthogonal) to the first component. Similarly, subsequent components capture the remaining variance.

Need for PCA

PCA is used to:

- Reduce the number of features in a dataset.
- Remove redundant or highly correlated information.
- Reduce computational complexity.
- Improve visualization of high-dimensional data.
- Reduce storage requirements.
- Help reduce the effect of noise.
- Make machine learning models faster and easier to interpret.
- Handle the **curse of dimensionality**.

Steps Involved in PCA

The main steps of PCA are:

1. **Collect the dataset:**
   Select the numerical features on which PCA will be applied.

2. **Standardize the data:**
   Features may have different scales, so they are usually standardized before applying PCA.

3. **Calculate the covariance matrix:**
   The covariance matrix helps identify how different features vary with respect to each other.

4. **Calculate eigenvalues and eigenvectors:**
   Eigenvectors determine the directions of the principal components, while eigenvalues indicate the amount of variance captured by each component.

5. **Select principal components:**
   Components with higher eigenvalues are generally selected because they contain more information.

6. **Transform the data:**
   The original features are projected onto the selected principal components, producing a new reduced dataset.

Principal Components

Suppose a dataset contains **10 original features**. Instead of using all 10 features, PCA may transform them into **2 or 3 principal components** while retaining a large percentage of the original information.

For example:

Original Features:

Age, Income, Spending, Savings, Expenses, ...

↓

PCA

↓

Principal Component 1, Principal Component 2

These components can then be used for visualization or as inputs to machine learning algorithms.

Explained Variance

Explained variance represents how much information or variation from the original dataset is retained by each principal component.

For example:

Component	Explained Variance
PC1	55%
PC2	25%
PC3	10%
PC4	5%

If PC1 and PC2 are selected, they together preserve **80% of the variance** in the original dataset.

Advantages of PCA

- Reduces the dimensionality of large datasets.
- Makes data visualization easier.
- Reduces computation time.
- Removes some redundancy from correlated features.
- Can improve the efficiency of machine learning algorithms.
- Helps in identifying the most significant directions of variation in data.

Limitations of PCA

- PCA can make the original features difficult to interpret because principal components are combinations of multiple features.
- It is sensitive to the scale of variables, so standardization is usually important.
- PCA mainly captures **linear relationships**.
- Some information may be lost when dimensions are reduced.

Applications of PCA

PCA is widely used in:

- Image processing and facial recognition.
- Data visualization.
- Pattern recognition.
- Feature extraction.
- Customer analysis.

- Medical data analysis.
- Text and document analysis.
- Machine learning preprocessing.

Conclusion:
PCA is an effective dimensionality reduction technique that converts a high-dimensional dataset into a smaller number of principal components while retaining as much important information as possible. It helps simplify datasets and makes subsequent machine learning analysis more efficient.

Code:

```
import numpy as np

import pandas as pd

from sklearn.preprocessing import StandardScaler

from sklearn.decomposition import PCA

import matplotlib.pyplot as plt

# Step 1: Load dataset (example: Iris dataset)

from sklearn.datasets import load_iris

data = load_iris()

X = data.data       # Features

y = data.target     # Labels

# Step 2: Standardize features

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

print("Explained variance ratio:", pca.explained_variance_ratio_)

print("Sum of explained variance:", sum(pca.explained_variance_ratio_))

plt.figure(figsize=(8,6))
```

```
for i, target_name in enumerate(data.target_names):
    plt.scatter(X_pca[y==i, 0], X_pca[y==i, 1], label=target_name)
plt.xlabel('PC1')
plt.ylabel('PC2')
plt.title('PCA of Iris Dataset')
plt.legend()
plt.show()
```

Explained variance ratio: [0.72962445 0.22850762]

Sum of explained variance: 0.9581320720000166

Output:

